

A PAPER WITHOUT AN LLM DISCLOSURE IS CAUSE FOR SUSPICION

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ABSTRACT. In contemporary mathematical and quantitative research, the absence of a disclosure regarding LLM use is cause for suspicion. If a large language model was used, the absence of a disclosure misrepresents how the research was done. If no large language model was used, the authors declined an inexpensive and effective check of their results without acknowledgment or justification. Thus a failure to disclose is evidence of either dishonesty or carelessness.

Introduction. In contemporary mathematical and quantitative research, a paper without an LLM disclosure is cause for suspicion. By an LLM disclosure, we mean a statement of whether or not large language models (LLMs) were used in conducting, checking, and/or writing up the research. The argument takes the form of a dilemma: either (i) an LLM was used in the research without disclosure, suggesting dishonesty or (ii) no LLM was used and no justification was offered, suggesting carelessness in verification. We consider these two cases in turn.

(i) Undisclosed use. Suppose that an LLM was used and the paper contains no LLM disclosure.¹ Standard publication ethics indicate a norm of disclosure [3]. arXiv expects disclosure of significant use [1], and SocArXiv requires disclosure of all use, including copyediting and background research [9]. One need not endorse every provision of these standards in order to recognize the expectation of disclosure they establish. Given that expectation, a failure to disclose LLM use is a misrepresentation of how the research was conducted. The omission gives readers reason to suspect dishonesty and to question the integrity of the work.

(ii) Non-use. We now suppose that no LLM was used and the paper contains no discussion of LLM use. LLM checks are inexpensive and can detect errors that would otherwise escape both the authors' own verification and peer review. Recent evidence includes checks of published economics replication packages, audits identifying errors in published education, economics, and mathematics papers, and benchmarks testing the detection of false mathematical claims [8, 2, 4, 5, 7].² Any proof or computation may contain errors, which is why verification is important. Authors are not obliged to perform every possible check. However, an author who declines an inexpensive and effective check should acknowledge that decision and report an explicit justification (e.g., confidentiality or ethical concerns). The omission gives readers reason to doubt the care taken over the results.

Conclusion. Our argument makes all contemporary mathematical and quantitative papers without an LLM disclosure suspect. We accept this consequence. If scholars are using LLMs, they should disclose it. If they are not, they should acknowledge and justify this decision.

Disclosure. This paper was written with the assistance of large language models, which included suggesting arguments, contributing to drafting and revision, and verification of arguments. The author holds that there are ethical concerns associated with the promotion of commercial large language models in mathematics, and thus chooses not to publicly identify the particular commercial tools used unless required for dissemination or publication. For this reason, the author expressly

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¹This is not hypothetical: recent trends in arXiv submissions strongly suggest high rates of undisclosed use [6].

²The author is in the process of completing and verifying an LLM-assisted audit of their complete body of work. Like Ferreira [4], the author has found that LLMs can be overzealous in flagging errors, but the associated burden is low for an author already familiar with the results.

departs from COPE’s requirement to identify particular tools [3]. Model details are available on request from p.aronow@yale.edu.

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